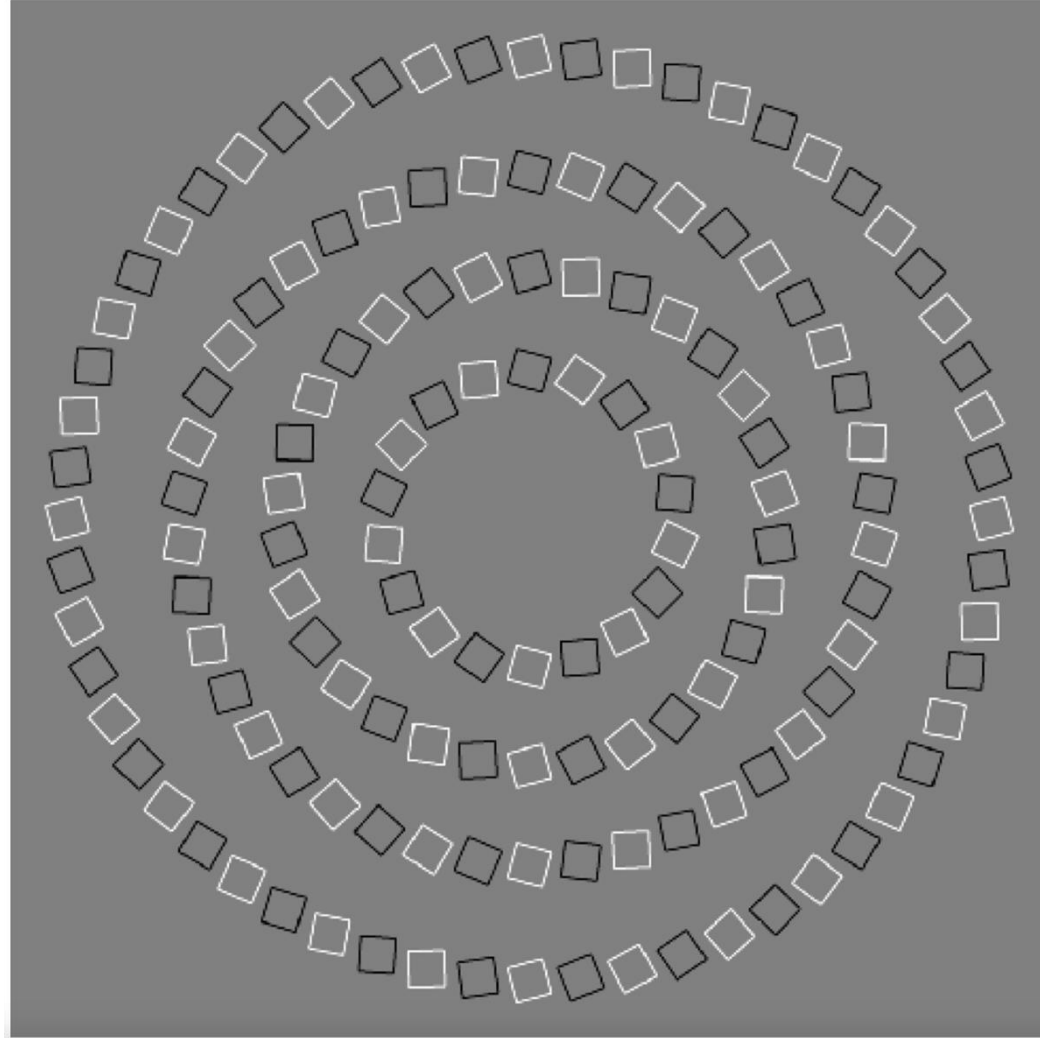


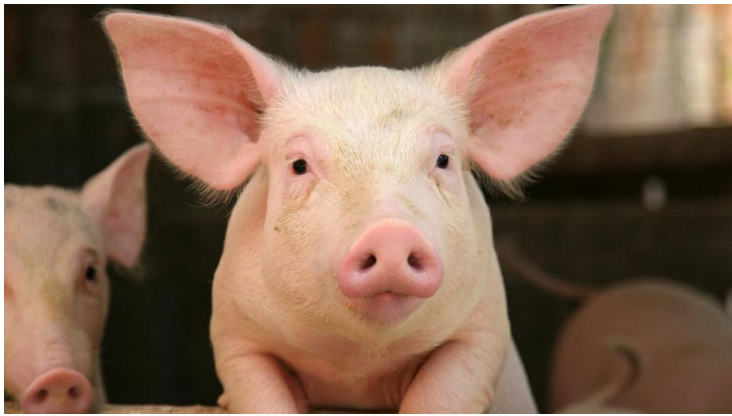
Adversarial Examples

Amit Shmidov | Tutorial 5

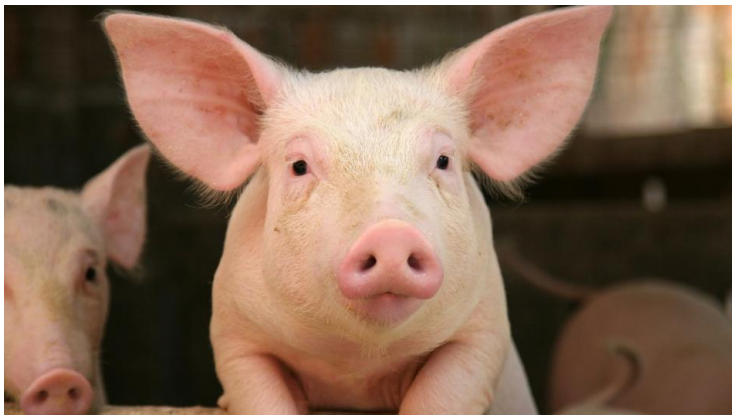
Largely relies on the corresponding presentation from DL4CV 2023, Weizmann Institute and a bit on the one from CS 231n 2017, Stanford University.

Any eyes can be tricked

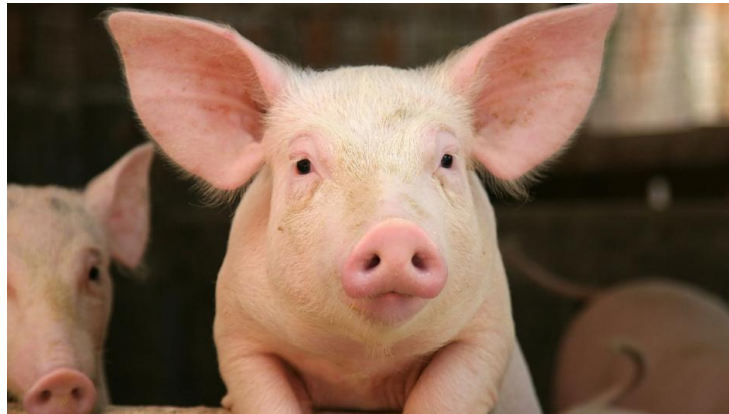




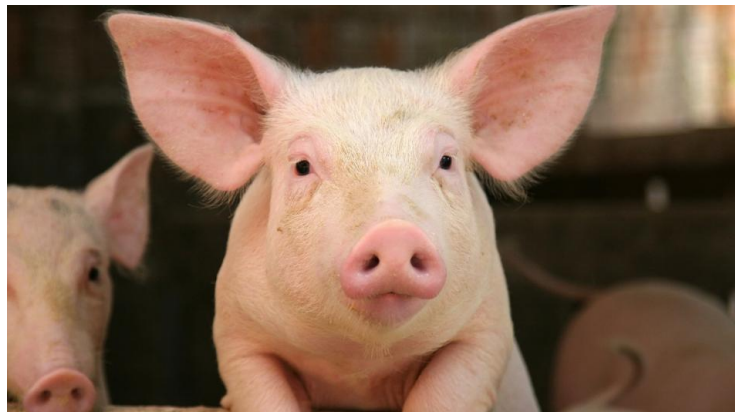
Pig (98.6%)



Pig (98.6%)



Airliner (99.0%)



+



$\times 0.02$

Pig (98.6%)

=



Airliner (99.0%)

Adversarial example?

Szegedy et al. 2013:

“We find that [by] applying an imperceptible non-random perturbation to a test image, it is possible to arbitrarily change the network’s prediction.

... we term the so perturbed examples ‘**adversarial examples**’”

Adversarial example?

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“We find that [by] applying an *imperceptible non-random* perturbation to a test image, it is possible to arbitrarily change the network’s prediction.

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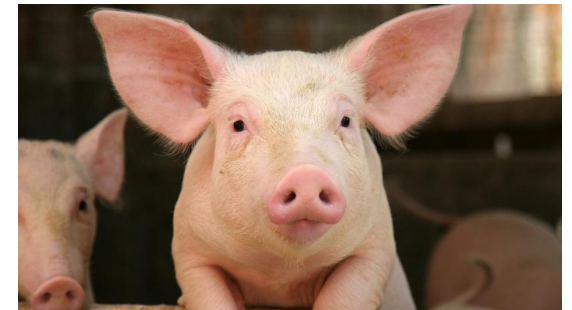


+



=

x 0.02

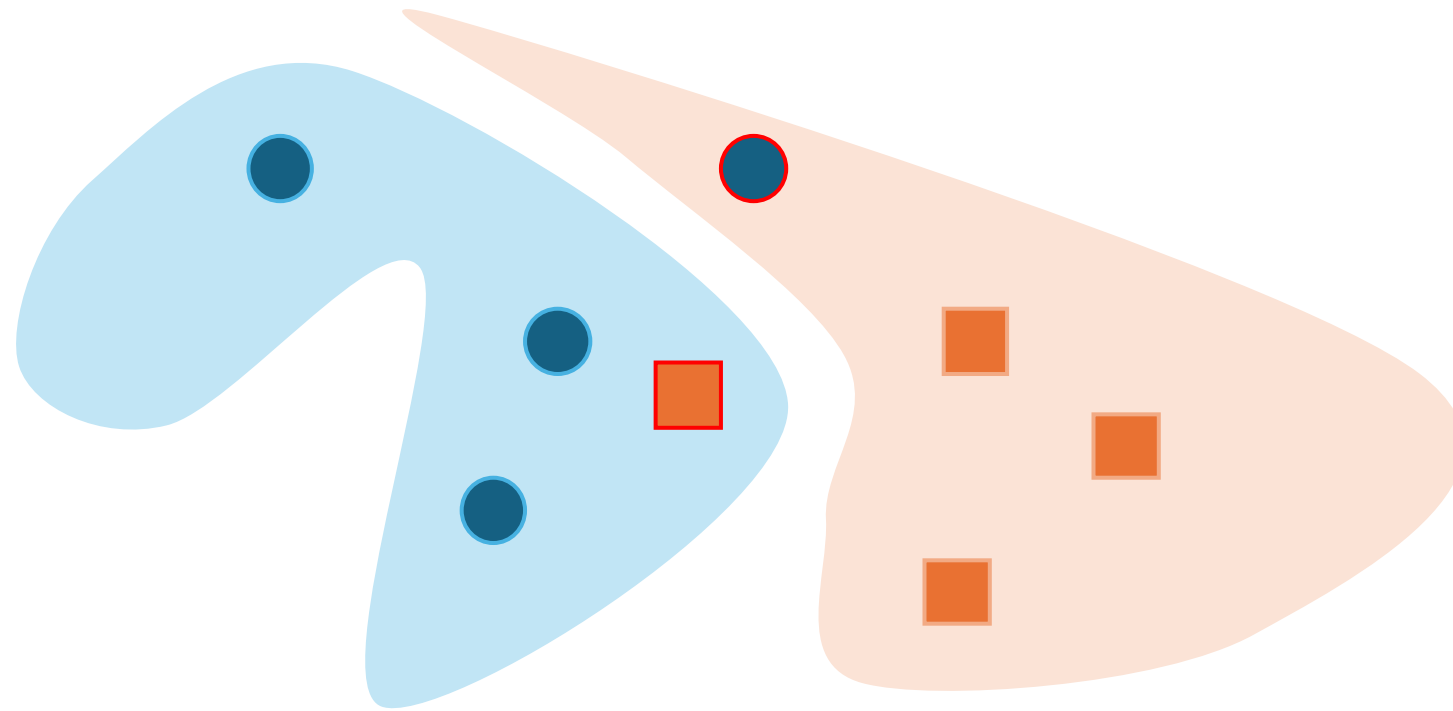


Not just for neural nets

- Linear models
 - Logistic regression
 - SVMs
- Decision trees
- Nearest neighbors

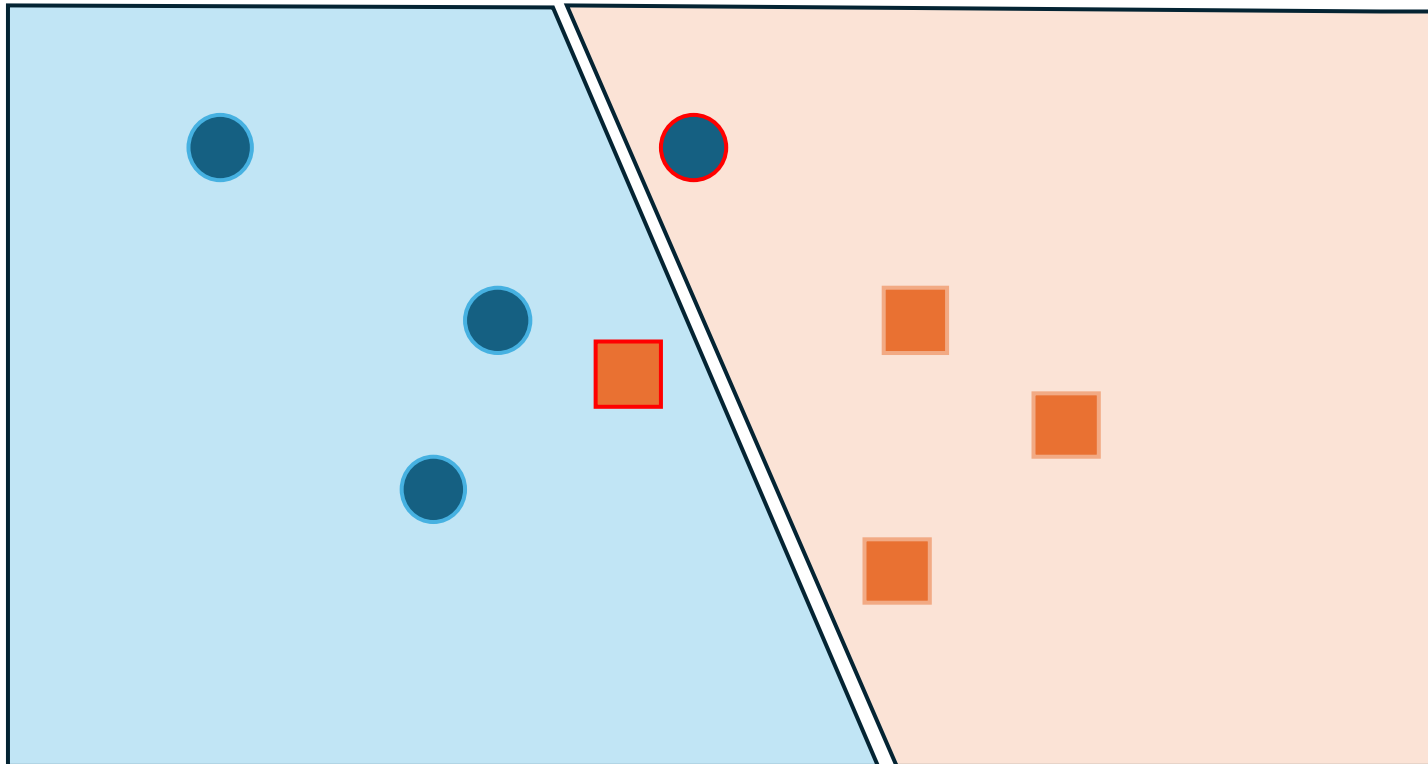
Not just for neural nets

Adversarial examples from overfitting



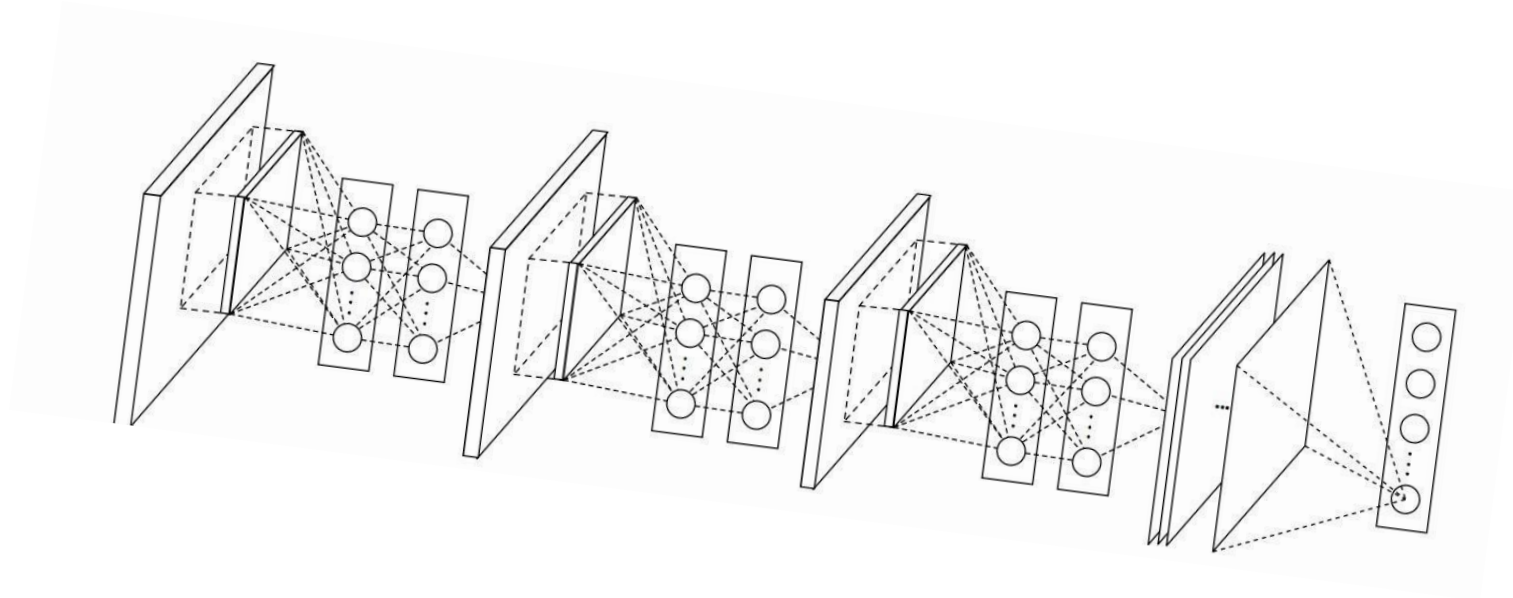
Not just for neural nets

Adversarial examples from linearity



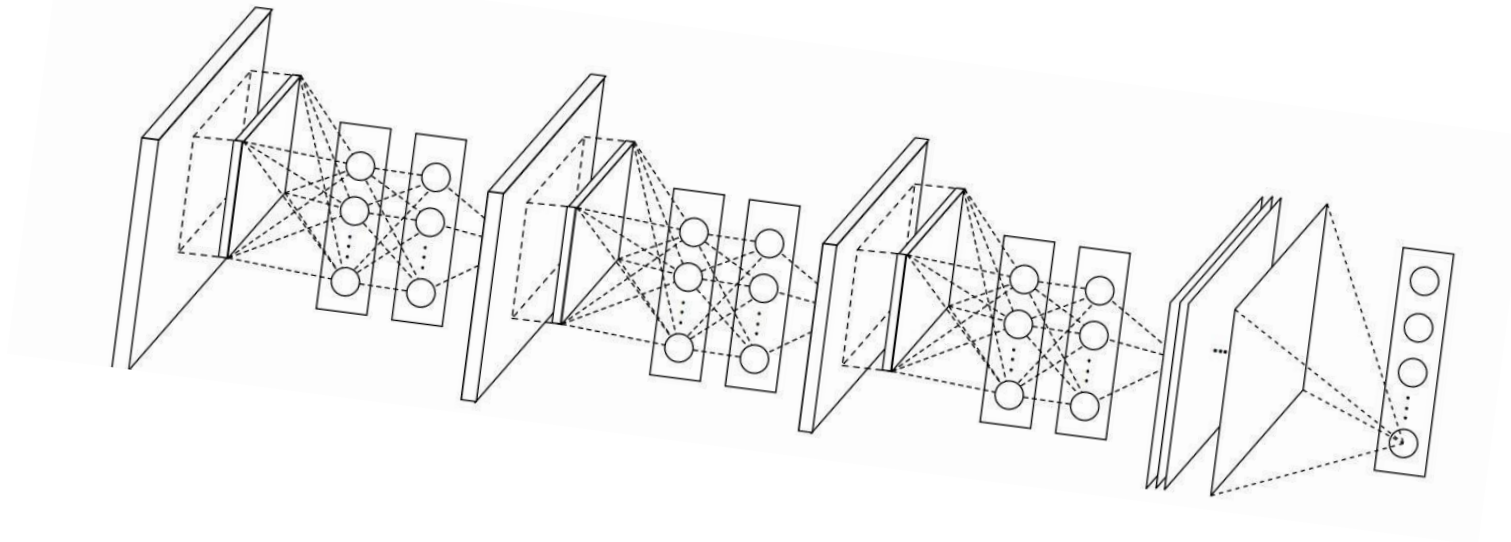
Recap on NN training

Recap on NN training



Recap on NN training

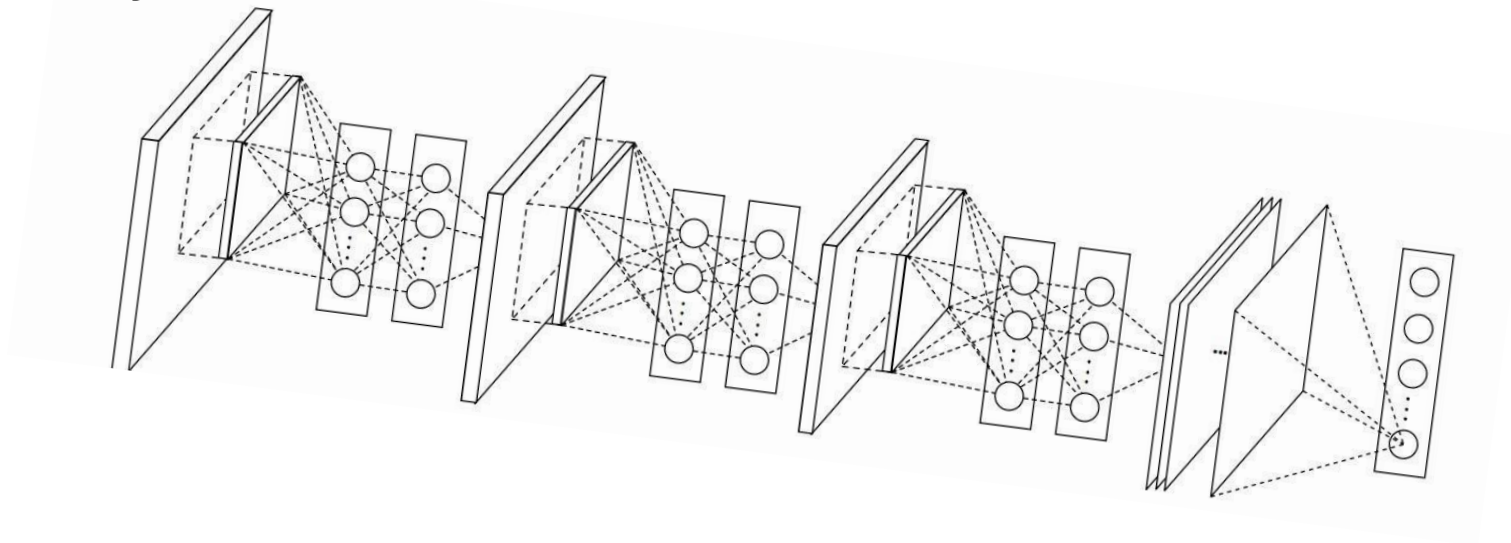
f_θ – NN-implemented function



$L(f_\theta(x), y)$ – Loss (how well we classify)

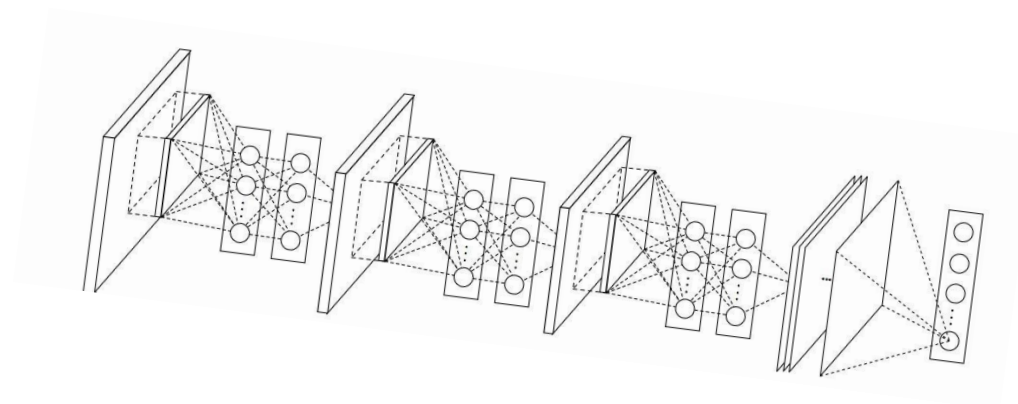
Recap on NN training

f_θ – NN-implemented function



$L(f_\theta(x), y)$ – Loss (how well we classify) $L_{\text{CrossEntropy}}(f_\theta(x), y) = -\log\left(\frac{e^{f_\theta(x)_y}}{\sum_j e^{f_\theta(x)_j}}\right)$

Recap on NN training



Training = loss minimization:

$$L(f_{\theta}(x), y) \rightarrow -\nabla_{\theta} L$$

Generating adversarial examples



Pig (89.7%)

$$f_{\theta}(x = \text{image}) = y = \text{pig}$$

Generating adversarial examples



Pig (89.7%)

$$f_{\theta}(x = \text{image}) \neq y = \text{pig}$$

Generating adversarial examples



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Let's shift x by some noise δ :

Generating adversarial examples



Pig (89.7%)

$$f_{\theta}(x = \text{image}) \neq y = \text{pig}$$

Let's shift x by some noise δ :

$$f_{\theta}(x + \delta) \neq y$$

Generating adversarial examples



Pig (89.7%)

$$f_{\theta}(x + \delta) \neq y$$

We want to measure how far δ takes $f_{\theta}(x + \delta)$ away from y

Generating adversarial examples



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and we already have the right measure for proximity: $L(f_{\theta}(x), y)$

Generating adversarial examples



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So, instead of minimizing L w.r.t θ , we could **maximize** it w.r.t δ (or x)!

Generating adversarial examples



Pig (89.7%)

$$f_{\theta}(x + \delta) \neq y$$

$$\delta = \nabla_x L$$

We want to measure how far δ takes $f_{\theta}(x + \delta)$ away from y
and we already have the right measure for proximity: $L(f_{\theta}(x), y)$
So, instead of minimizing L w.r.t θ , we could **maximize** it w.r.t δ (or x)!

Generating adversarial examples



x : Pig (89.7%)



$x + \nabla_x L$: Hay (68.8%)



$x + 10\nabla_x L$: Pig (44.7%)



$x + 100\nabla_x L$: Fireguard (44.8%)

Generating adversarial examples



$x + 1000\nabla_x L$: Spotlight (99.9%)

Generating adversarial examples



$x + 1000\nabla_x L$: Spotlight (99.9%)

What do you see in the image?
Is it really an adversarial example?

Generating adversarial examples



$\neq$



$x + 1000\nabla_x L$: Spotlight (99.9%)

x : Airliner (99.0%)

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Is it really an adversarial example?

Generating adversarial examples



$\neq$



$x + 1000\nabla_x L$: Spotlight (99.9%)

x : Airliner (99.0%)

What do you see in the image?
Is it really an adversarial example?

We need a **small δ**

Generating adversarial examples

We need a **small** δ

$$\|\delta\| \leq \varepsilon$$

Generating adversarial examples

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δ is in the size of x , so we can decide on:

$$\|\delta\|_{\infty} = \max \delta_{i,j} \leq \varepsilon$$

Generating adversarial examples

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δ is in the size of x , so we can decide on:

$$\|\delta\|_{\infty} = \max \delta_{i,j} \leq \varepsilon$$

$\varepsilon = 0.1$:

$$\begin{pmatrix} 0.09 & -0.065 & \cancel{0.2} \\ 0.04 & 0.05 & -0.02 \\ -7 \cdot 10^{-6} & -0.009 & 0.024 \end{pmatrix}$$

Generating adversarial examples

We demand a small δ and $\|\delta\|_\infty \leq \varepsilon$

Generating adversarial examples

Let's enforce $g(\|\nabla_x L\|) \leq \varepsilon$

Generating adversarial examples

Let's enforce $g(\|\nabla_x L\|) \leq \varepsilon$:

$$\begin{pmatrix} 0.09 & -0.065 & 0.2 \\ 0.04 & 0.05 & -0.02 \\ -7 \cdot 10^{-6} & -0.009 & 0.024 \end{pmatrix} \quad \delta = \quad \nabla_x L$$

Generating adversarial examples

Let's enforce $g(\|\nabla_x L\|) \leq \varepsilon$:

$$\begin{pmatrix} 1 & -1 & 1 \\ 1 & 1 & -1 \\ -1 & -1 & 1 \end{pmatrix} \quad \delta = \quad \text{sgn}(\nabla_x L)$$

Generating adversarial examples

Let's enforce $g(\|\nabla_x L\|) \leq \varepsilon$:

$$\begin{pmatrix} \varepsilon & -\varepsilon & \varepsilon \\ \varepsilon & \varepsilon & -\varepsilon \\ -\varepsilon & -\varepsilon & \varepsilon \end{pmatrix}$$

$$\delta = \varepsilon \cdot \text{sgn}(\nabla_x L)$$

Generating adversarial examples

Let's enforce $g(\|\nabla_x L\|) \leq \varepsilon$:

$$\|\delta\| = \|\varepsilon \cdot \text{sgn}(\nabla_x L)\| \leq \varepsilon$$

Generating adversarial examples

$$\delta = \varepsilon \cdot \text{sgn}(\nabla_x L)$$

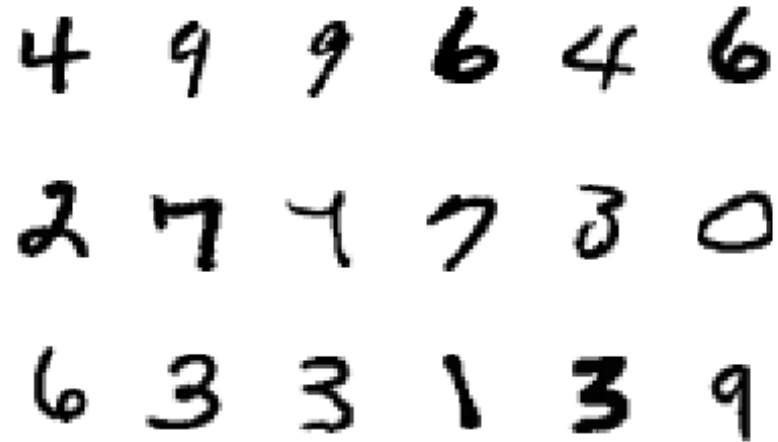
Fast Gradient Sign Method (FGSM)

Goodfellow et al. 2015

$$\delta = \varepsilon \cdot \text{sgn}(\nabla_x L)$$

Fast Gradient Sign Method (FGSM)

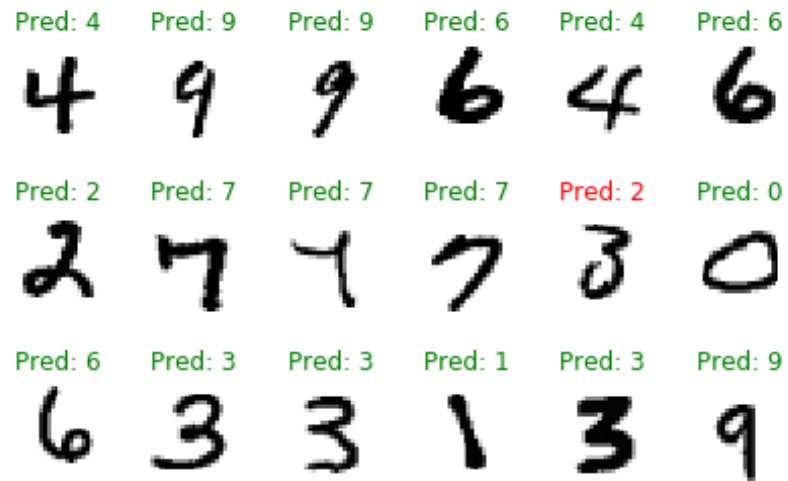
Goodfellow et al. 2015



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Fast Gradient Sign Method (FGSM)

Goodfellow et al. 2015



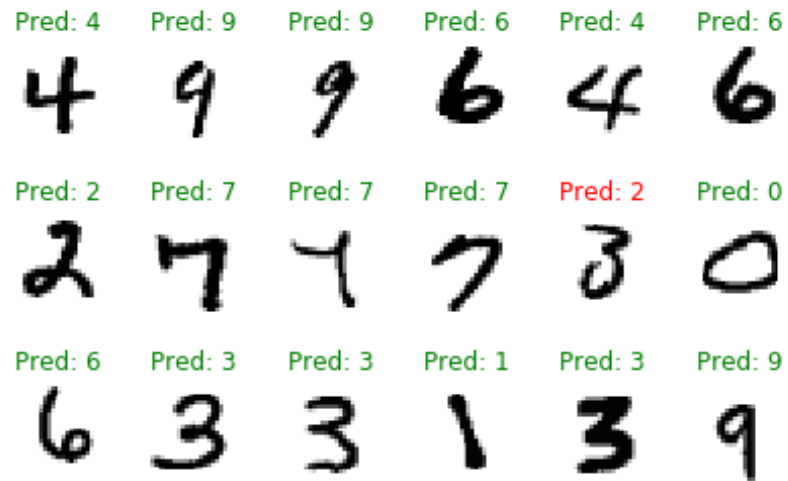
Classifier:

```
model = nn.Sequential(
    nn.Conv2d(1, 16, 4, stride=2, padding=1),
    nn.ReLU(),
    nn.Conv2d(16, 32, 4, stride=2, padding=1),
    nn.ReLU(),
    Flatten(),
    nn.Linear(32 * 7 * 7, 100),
    nn.ReLU(),
    nn.Linear(100, 10)
)
```


$$\delta = \varepsilon \cdot \text{sgn}(\nabla_x L)$$

Fast Gradient Sign Method (FGSM)

Goodfellow et al. 2015

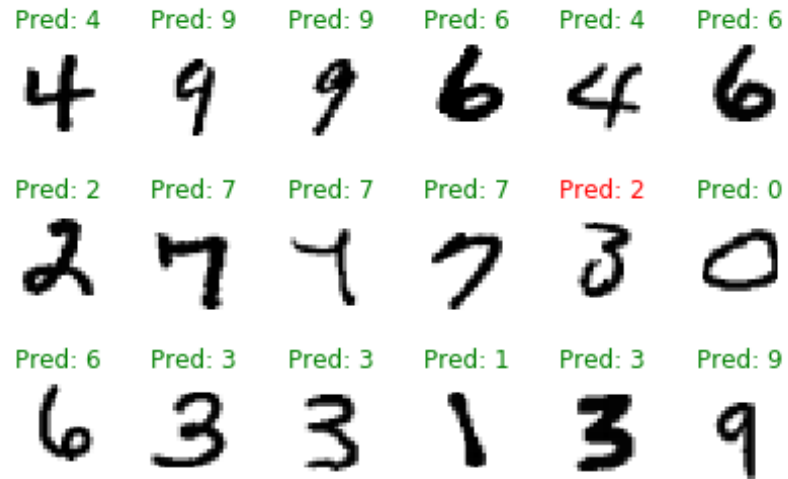


$$X_{adv} = X + \varepsilon \cdot \text{sgn}(\nabla_x L(f_\theta(X), Y))$$

$$\delta = \varepsilon \cdot \text{sgn}(\nabla_x L)$$

Fast Gradient Sign Method (FGSM)

Goodfellow et al. 2015





$$X_{adv} = X + \varepsilon \cdot \text{sgn}(\nabla_X L(f_\theta(X), Y))$$

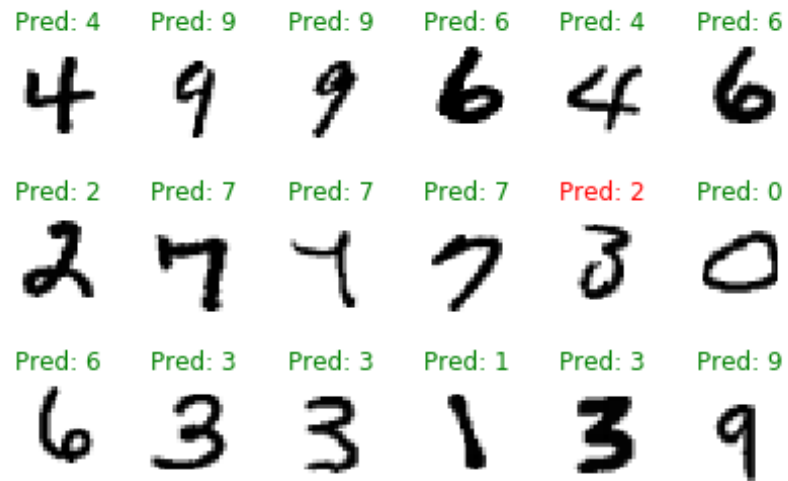
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Fast Gradient Sign Method (FGSM)

Goodfellow et al. 2015



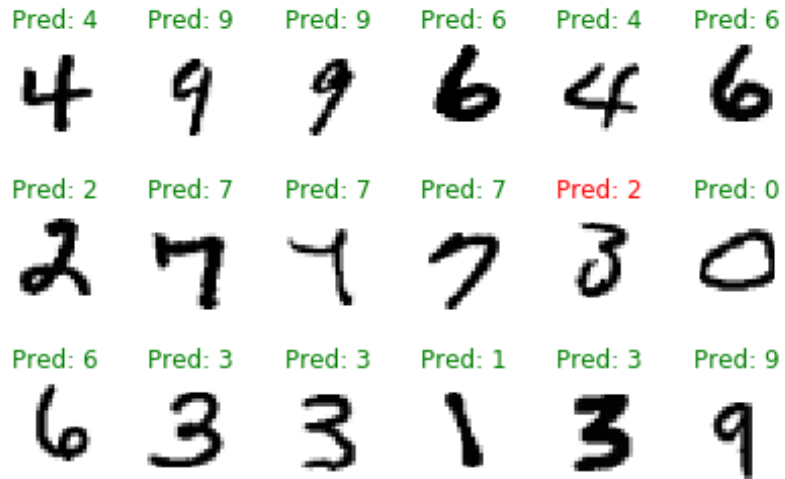
$$X_{adv} = X + \varepsilon \cdot \text{sgn}(\nabla_x L(f_\theta(X), Y))$$



$$\delta = \varepsilon \cdot \text{sgn}(\nabla_x L)$$

Fast Gradient Sign Method (FGSM)

Goodfellow et al. 2015



$$X_{adv} = X + \varepsilon \cdot \text{sgn}(\nabla_x L(f_\theta(X), Y))$$



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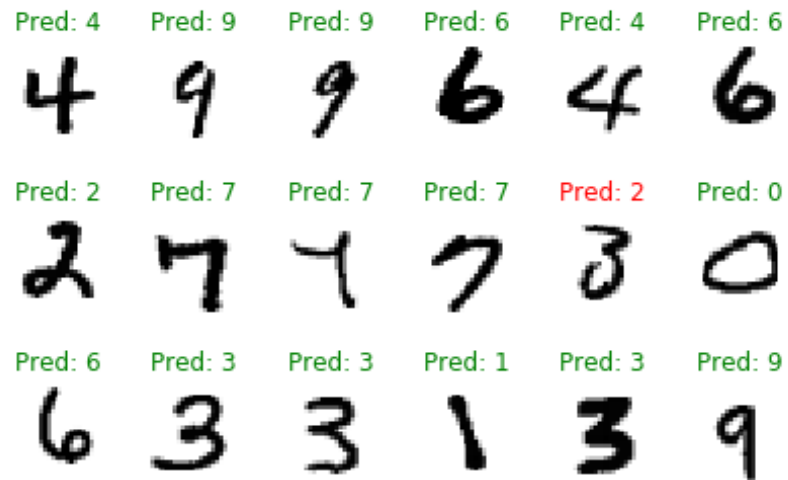
Fast Gradient Sign Method (FGSM)

Goodfellow et al. 2015

Test success rate: 98.7%
FGSM success rate: 40.0%



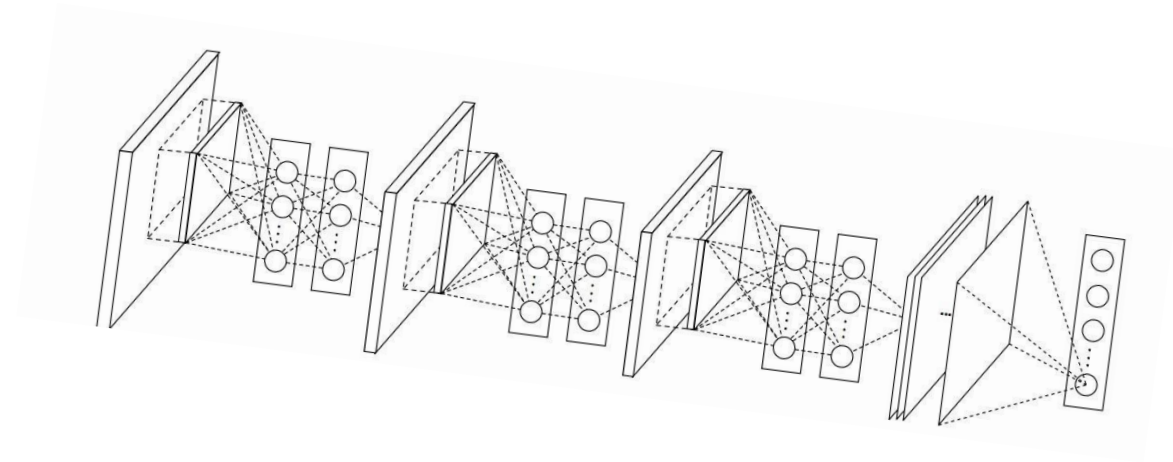
$$X_{adv} = X + \varepsilon \cdot \text{sgn}(\nabla_x L(f_\theta(X), Y))$$



How to defend?

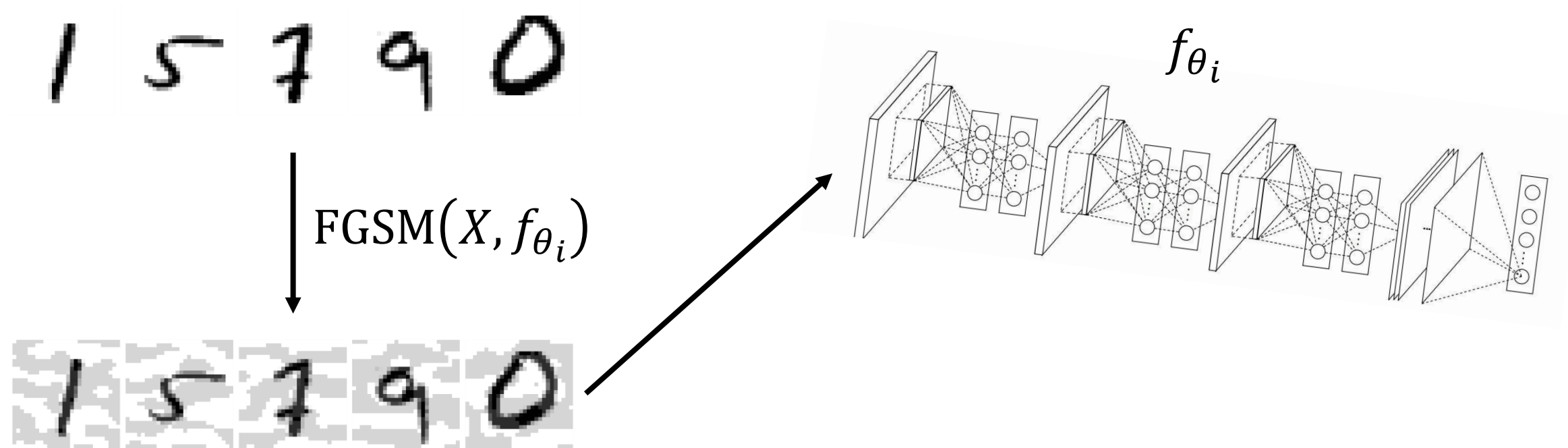
How to defend?

Train on adversarial examples:



How to defend?

Train on adversarial examples:



Adversarial training

Method	Success Rate
Standard training	98.7

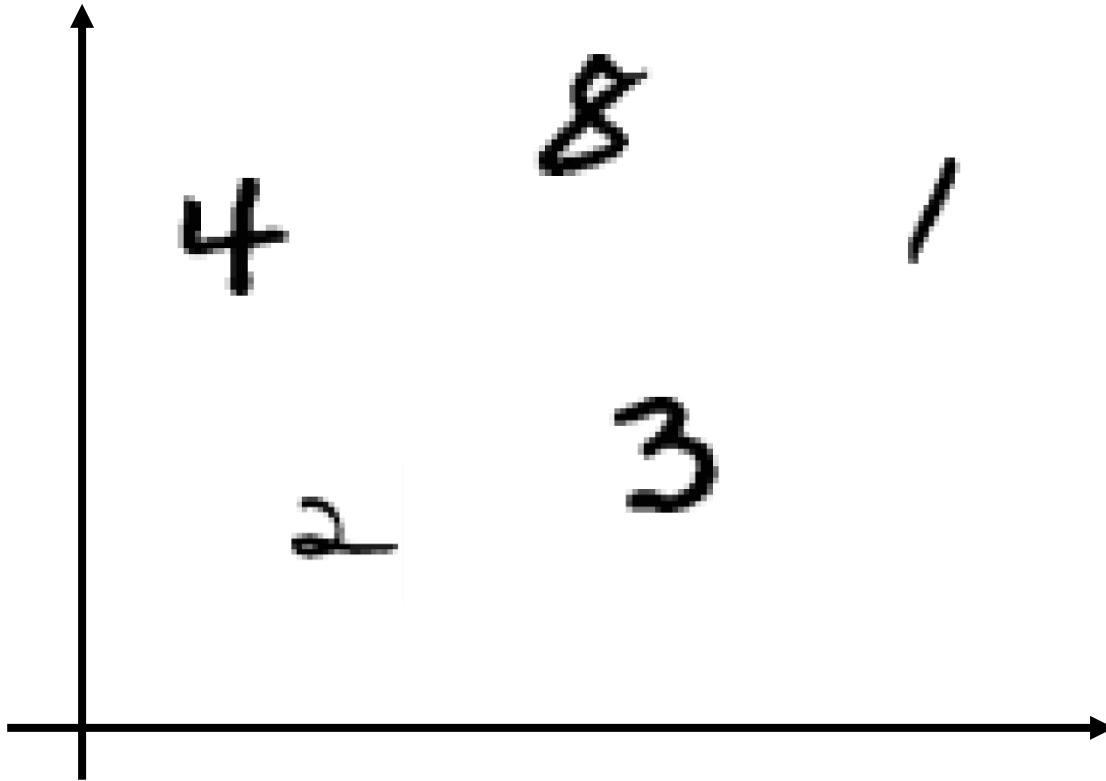
Adversarial training

Method	Success Rate	FGSM
Standard training	98.7	40.0

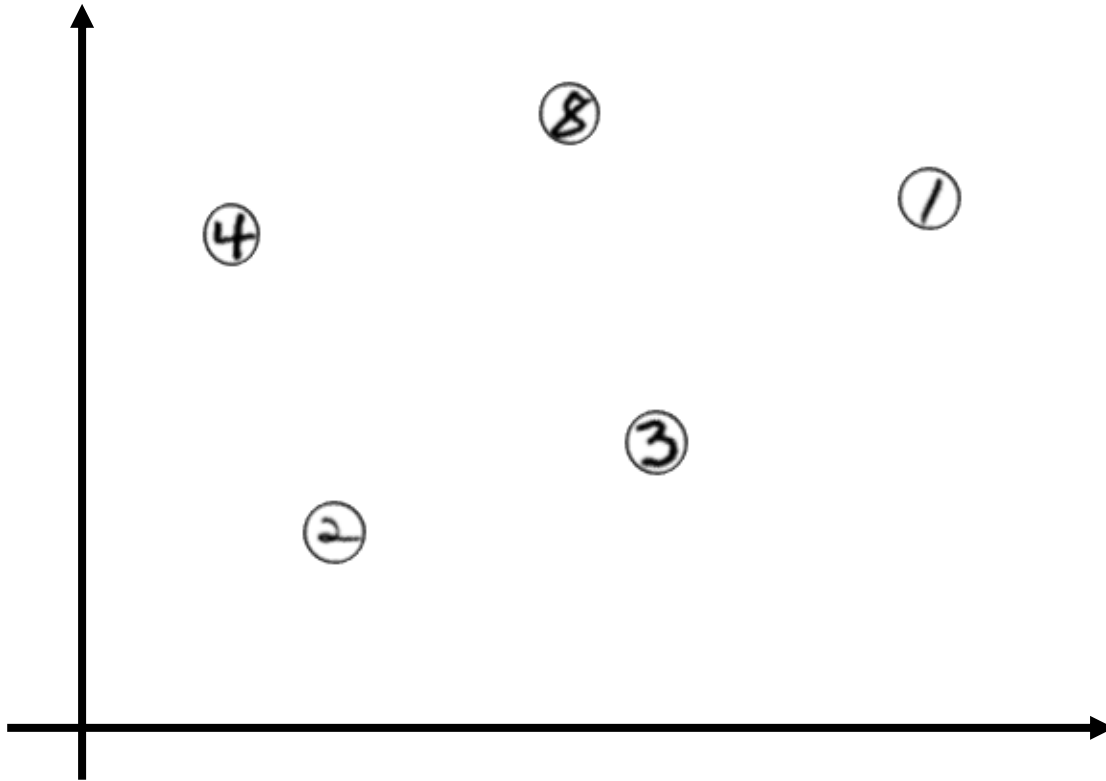
Adversarial training

Method	Success Rate	FGSM
Standard training	98.7	40.0
FGSM adv. training	97.2	94.0

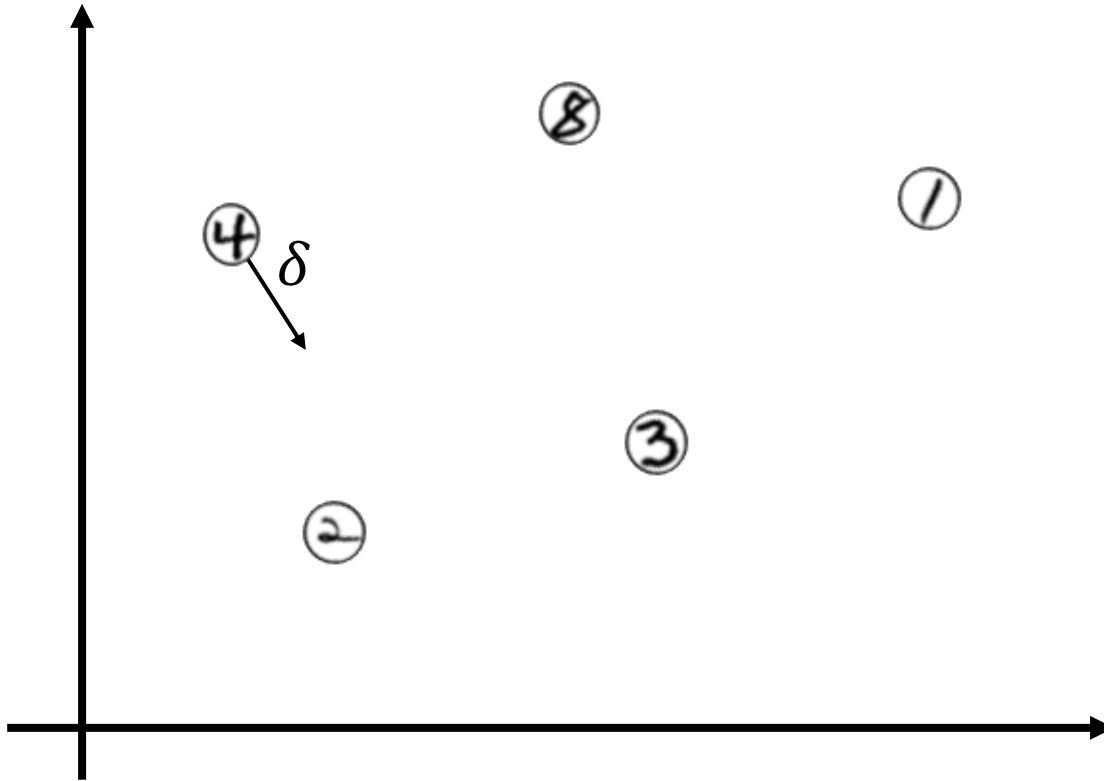
Perturbation attack – graphic view



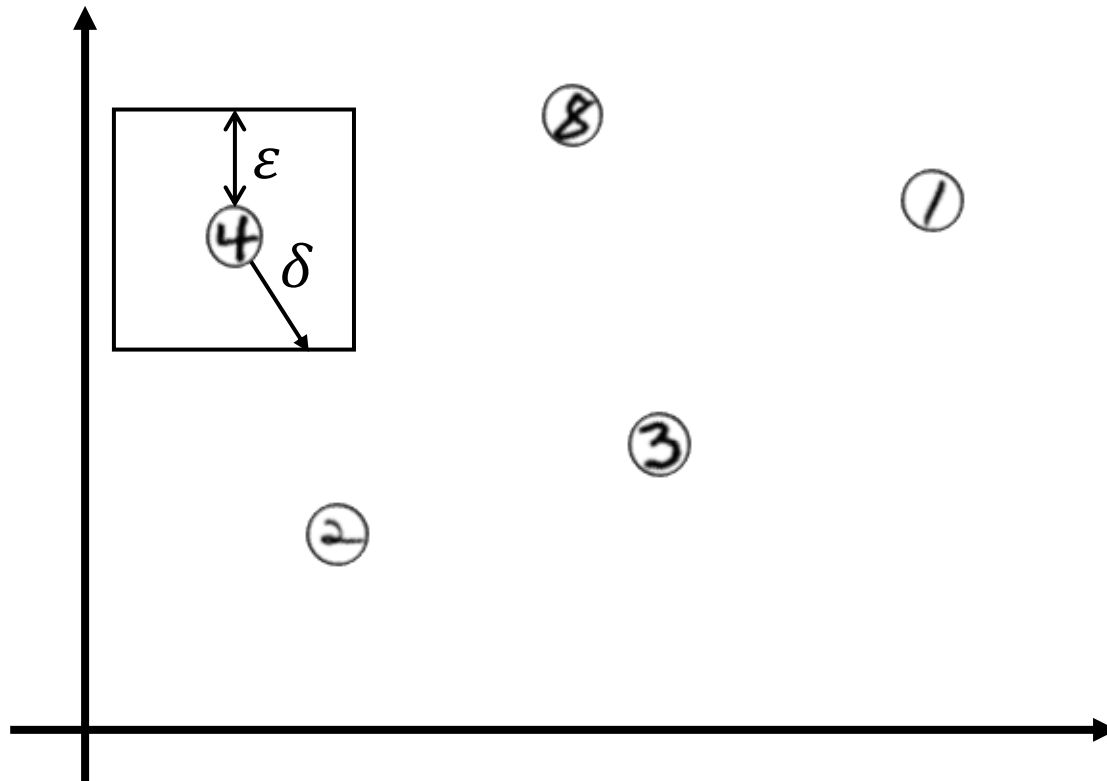
Perturbation attack – graphic view



Perturbation attack – graphic view

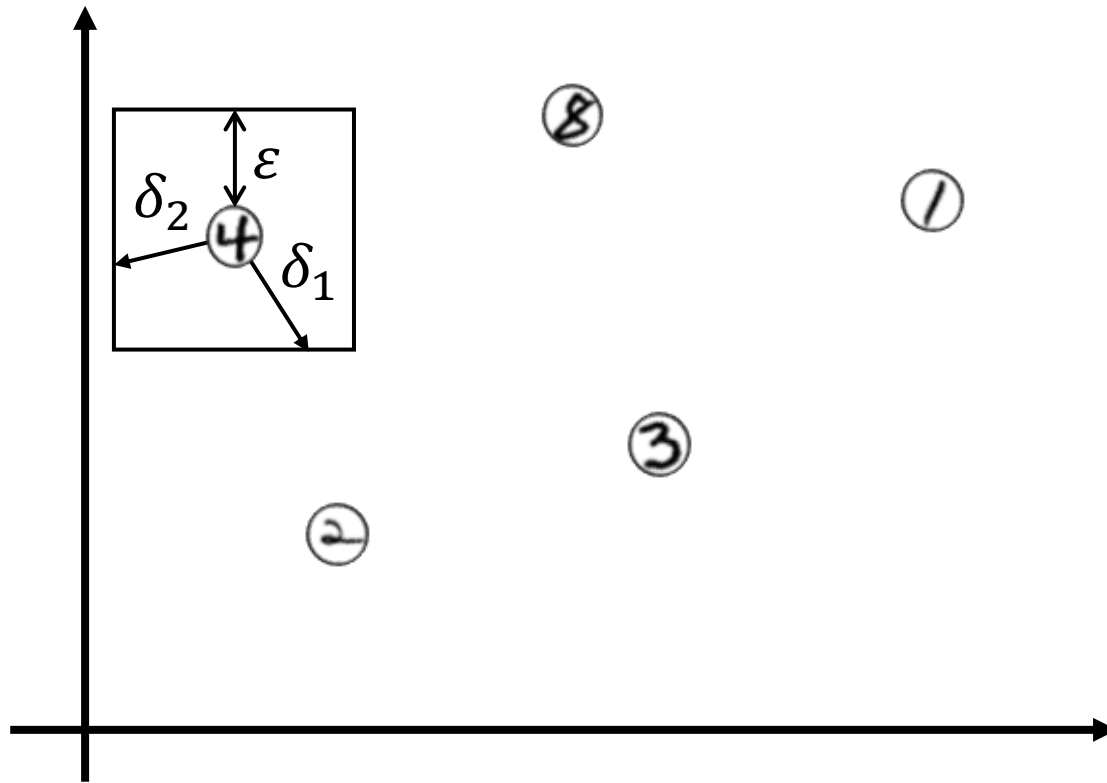


Perturbation attack – graphic view



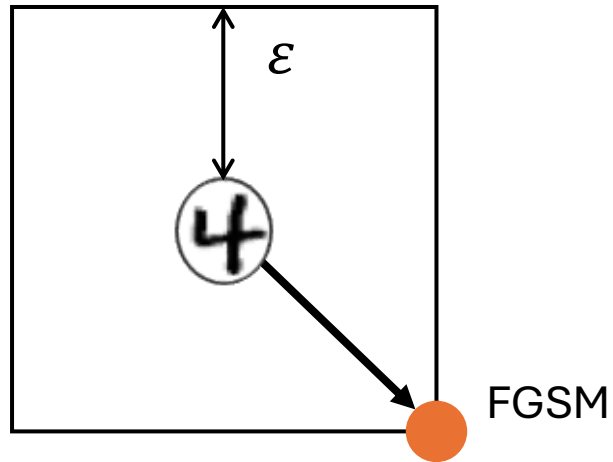
$$\|\delta\|_{\infty} \leq \epsilon$$

Perturbation attack – graphic view

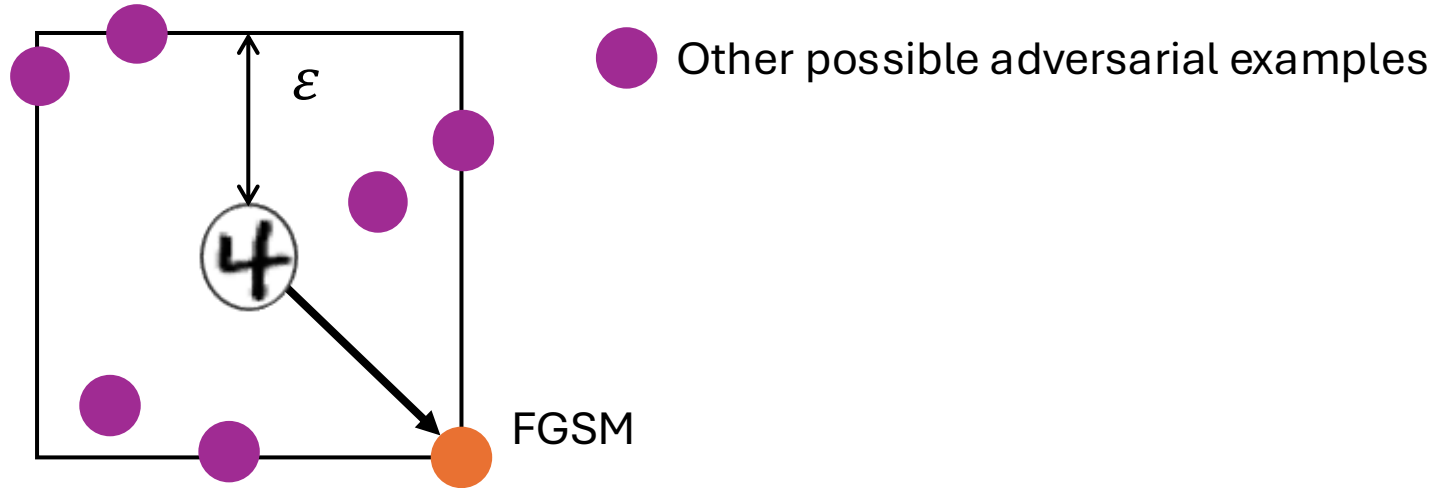


$$\|\delta\|_{\infty} \leq \epsilon$$

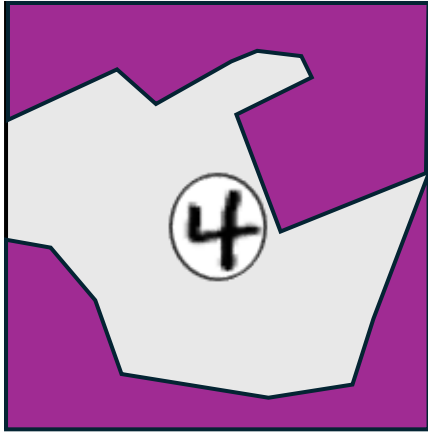
Perturbation attack – graphic view



Perturbation attack – graphic view



Perturbation attack – graphic view



● Other possible adversarial examples

Projected Gradient Descent (PGD)

Attack model:

$$s = \{\delta \mid \|\delta\|_{\infty} \leq \varepsilon\}$$

FGSM:

$$X_{adv} = X + \varepsilon \cdot \text{sgn}(\nabla_X L(f_{\theta}(X), Y))$$

Projected Gradient Descent (PGD)

Attack model:

$$s = \{\delta \mid \|\delta\|_\infty \leq \varepsilon\}$$

FGSM:

$$X_{adv} = X + \varepsilon \cdot \text{sgn}(\nabla_X L(f_\theta(X), Y))$$

PGD (=iterative GSM):

run GSM N times.

Projected Gradient Descent (PGD)

PGD (=iterative GSM):

$$X_{adv}^0 = X$$

$$X_{adv}^{N+1} = X_{adv}^N + \alpha \cdot \text{sgn} \left(\nabla_X L \left(f_{\theta}(X_{adv}^N, Y_{true}) \right) \right)$$

Projected Gradient Descent (PGD)

PGD (=iterative GSM):

$$X_{adv}^0 = X$$

$$X_{adv}^{N+1} = \text{Clip}_{X,\varepsilon} \left[X_{adv}^N + \alpha \cdot \text{sgn} \left(\nabla_X L \left(f_\theta(X_{adv}^N, Y_{true}) \right) \right) \right]$$



Projected Gradient Descent (PGD)

Method	Success Rate	FGSM	PGD
Standard training	98.7	40.0	7.3
FGSM adv. training	97.2	94.0	90.0

Projected Gradient Descent (PGD)

Method	Success Rate	FGSM	PGD
Standard training	98.7	40.0	7.3
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PGD adv. training	98.0	96.1	95.9

Projected Gradient Descent (PGD)

Method	Success Rate	FGSM	PGD
Standard training	98.7	40.0	7.3
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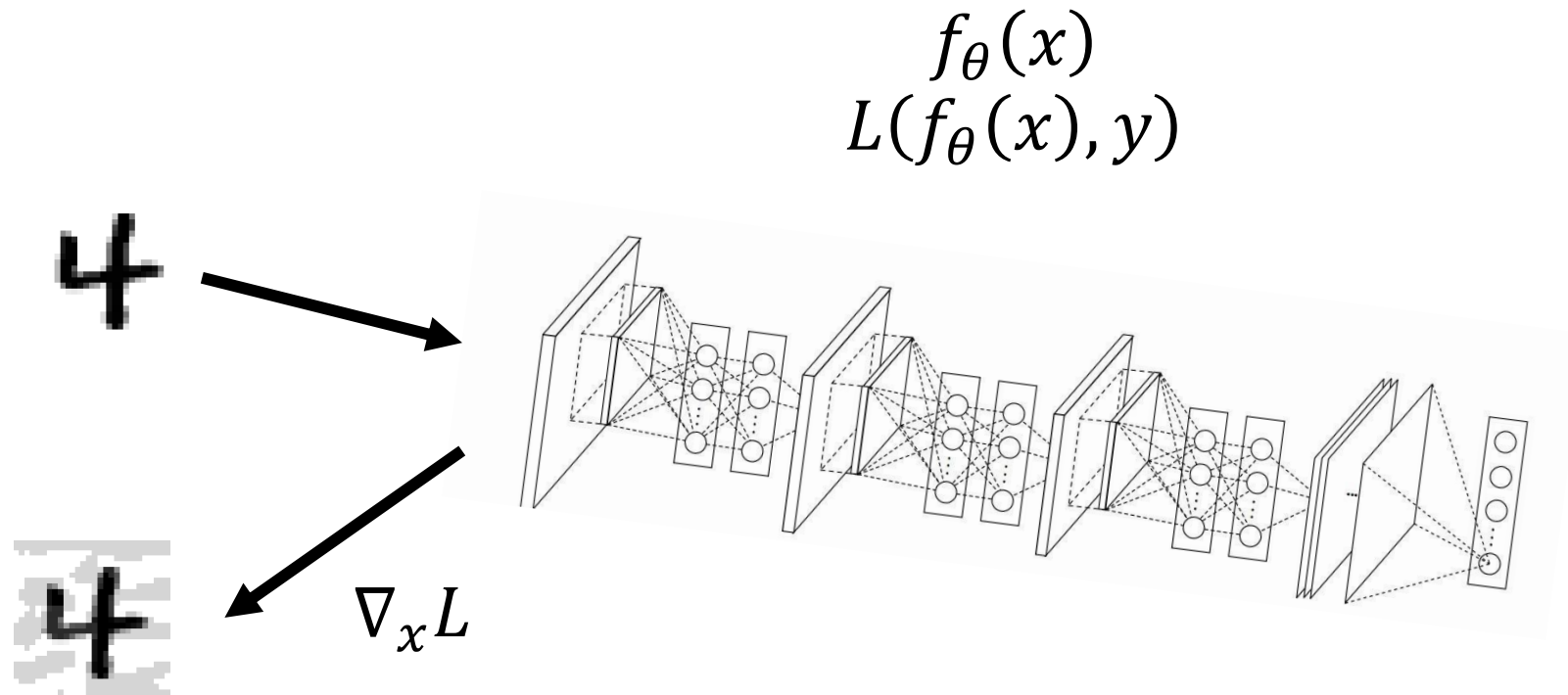
But what about other datasets?

CIFAR10 (ResNet50)	Success Rate	PGD ($\epsilon = 8/255$)
Standard training	95.25	0.0
PGD adv. training	87.03	53.29

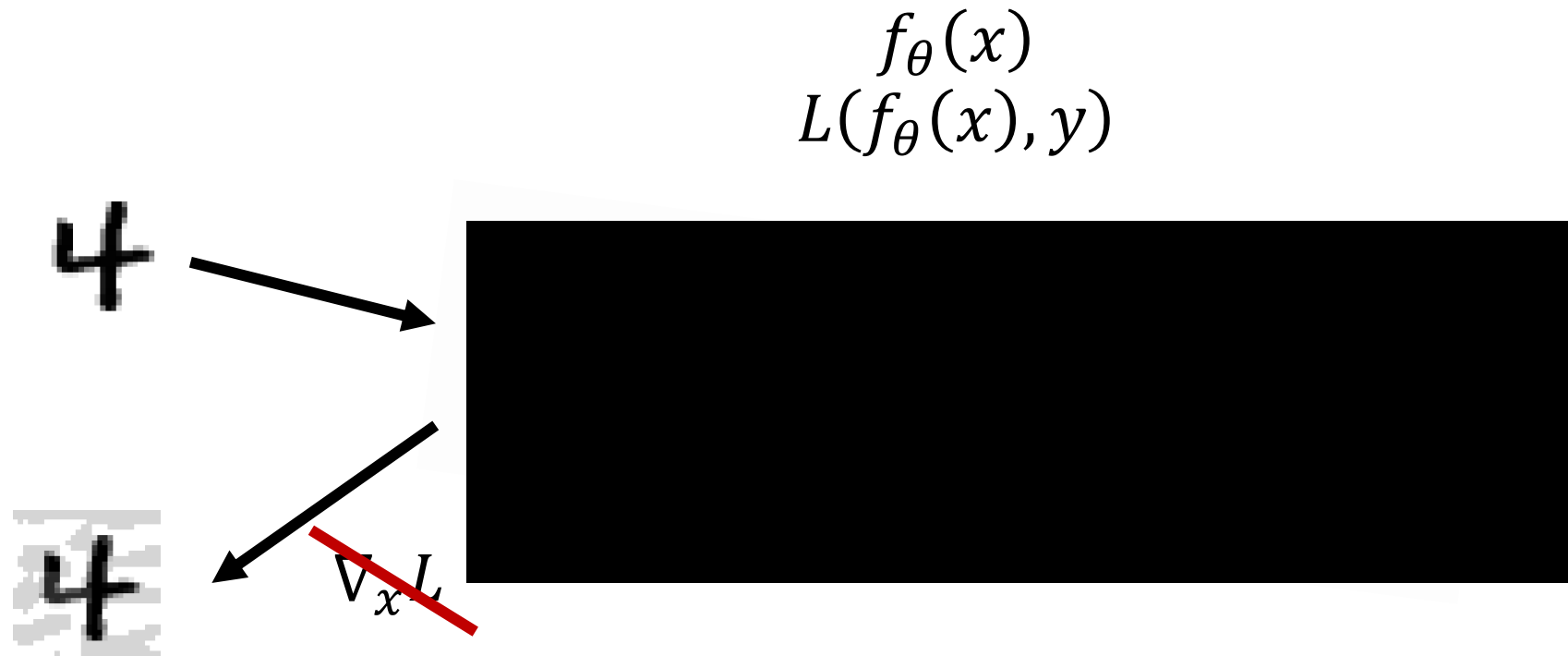
ImageNet (ResNet50)	Success Rate	PGD ($\epsilon = 8/255$)
Standard training	76.13	0.01
PGD adv. training	47.91	19.52

Source: <https://github.com/MadryLab/robustness>

Black-box attacks



Black-box attacks



Black-box attacks – transferability

	ResNet-152	ResNet-101	ResNet-50	VGG-16	GoogleNet
	91	91.7	92.1	89	88.3
	ResNet-152	ResNet-101	ResNet-50	VGG-16	GoogleNet
ResNet-152	32				
ResNet-101		33			
ResNet-50			29		
VGG-16				5	
GoogleNet					11

Top-5 accuracy on ImageNet

Liu et al. 2016, “Delving into Transferable Adversarial Examples and Black-box Attacks”, Table 20

Black-box attacks – transferability

	ResNet-152	ResNet-101	ResNet-50	VGG-16	GoogleNet
	91	91.7	92.1	89	88.3
	ResNet-152	ResNet-101	ResNet-50	VGG-16	GoogleNet
ResNet-152		55	53	47	36
ResNet-101	56		50	46	40
ResNet-50	59	53		47	38
VGG-16	42	39	41		21
GoogleNet	71	74	62	53	

Top-5 accuracy on ImageNet

Liu et al. 2016, “Delving into Transferable Adversarial Examples and Black-box Attacks”, Table 20

Black-box attacks – transferability

	ResNet-152	ResNet-101	ResNet-50	VGG-16	GoogLeNet
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Why does it work?

Black-box attacks – transferability

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ResNet-152		55	53	47	36
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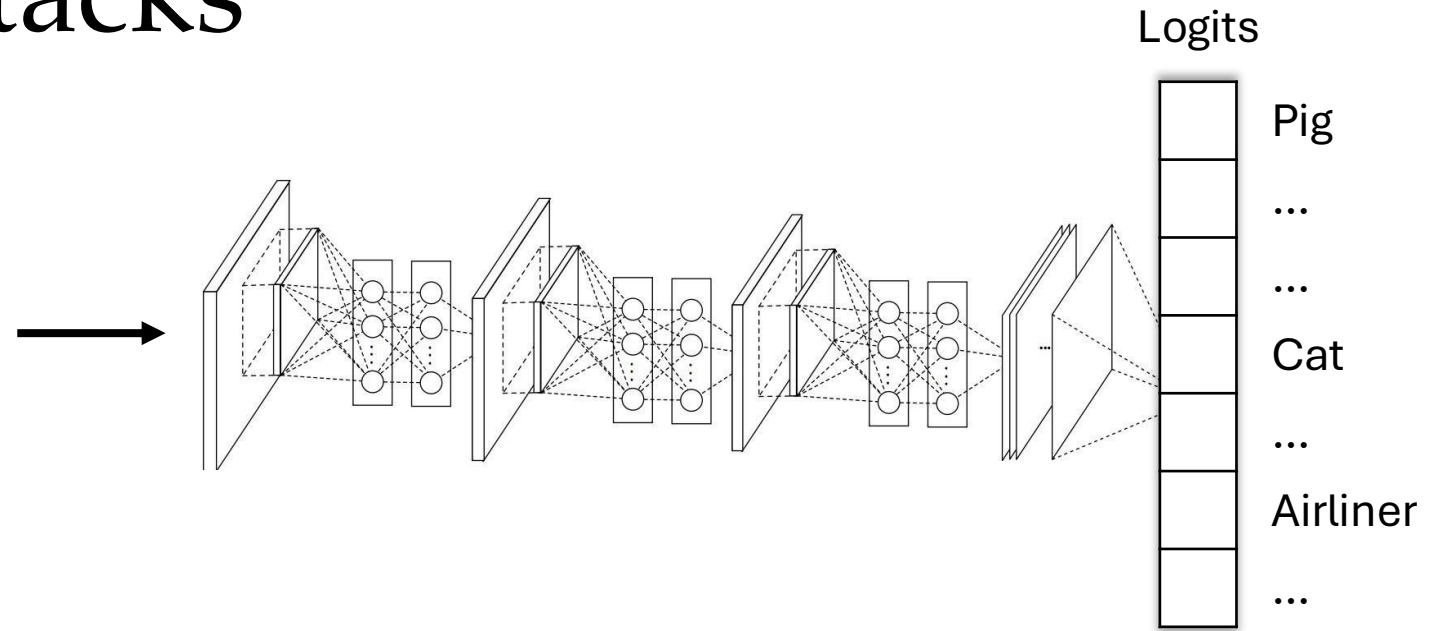
Why does it work?

1. Adversarial examples come from the data
2. Models have similar backbones

Targeted attacks



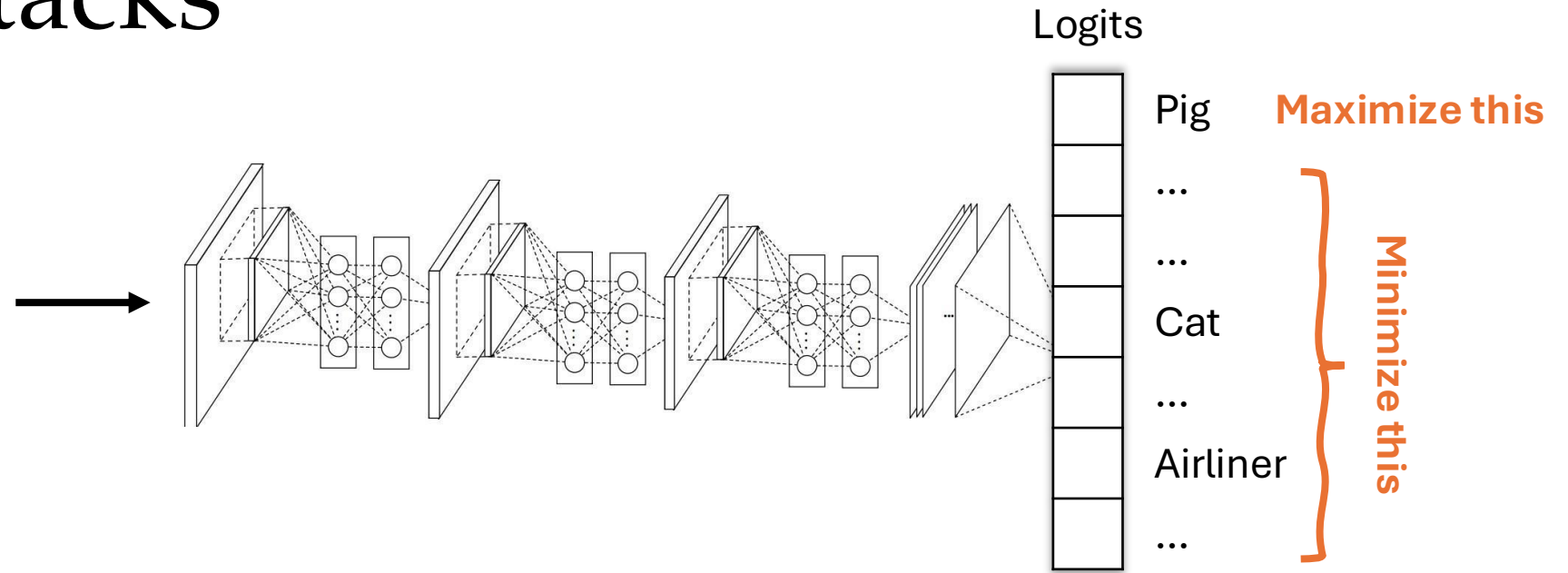
x : Pig (89.7%)



Targeted attacks



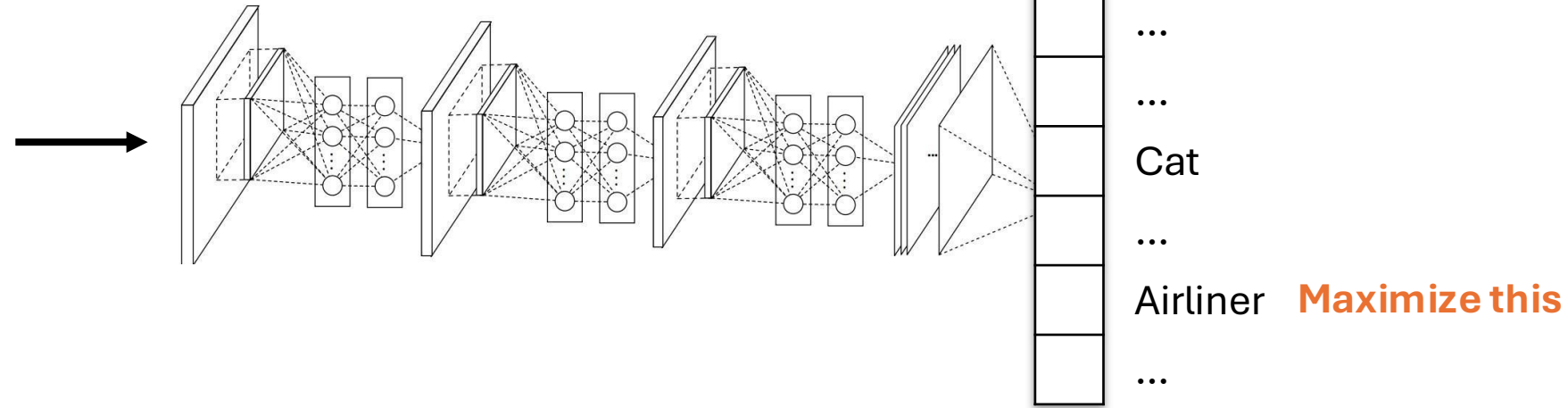
x : Pig (89.7%)



Targeted attacks



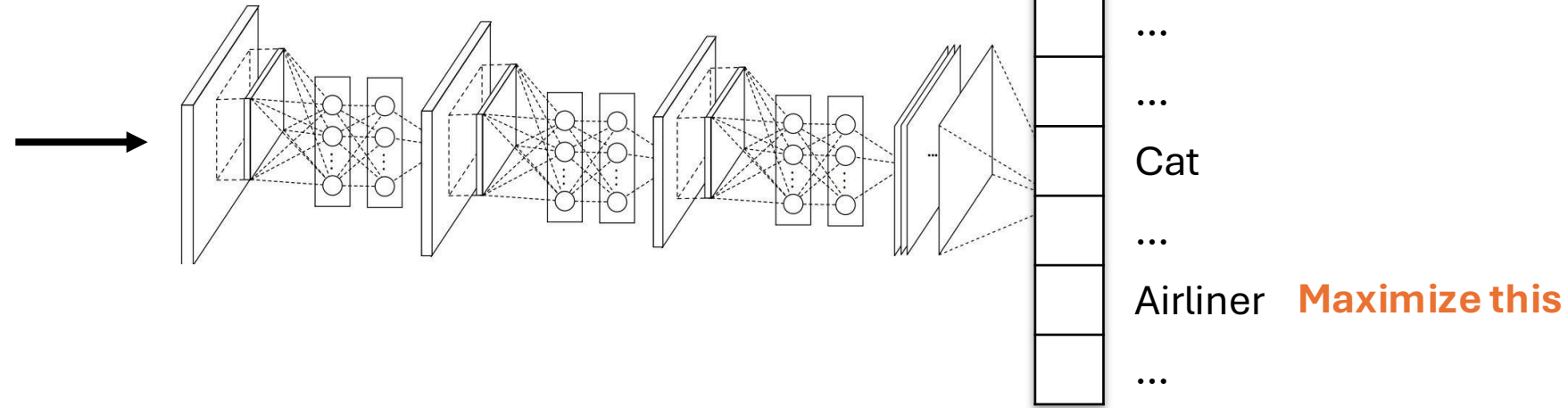
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Targeted attacks



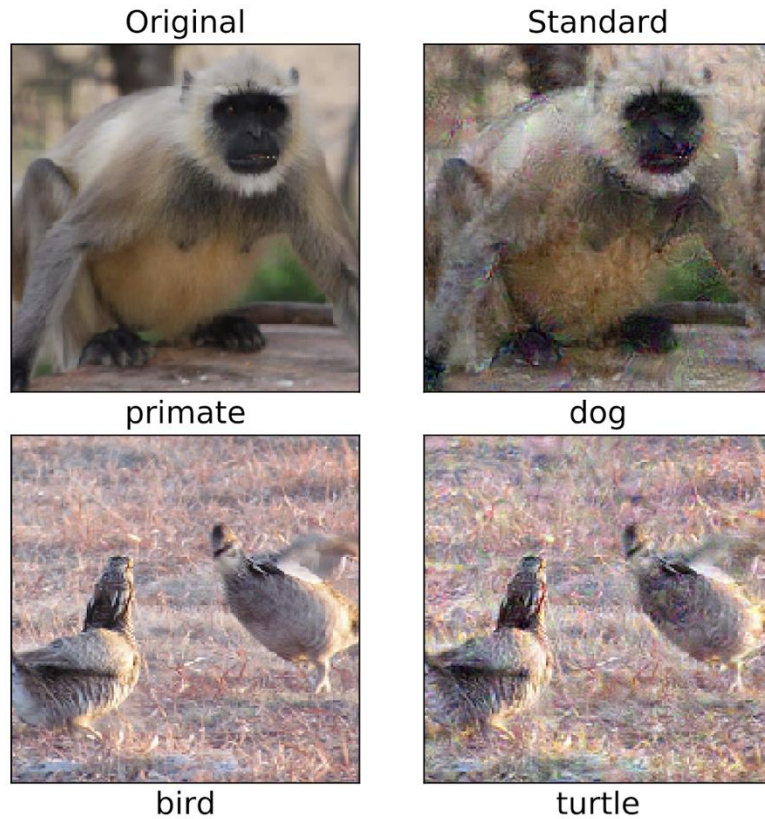
x : Pig (89.7%)



$$\nabla_x L(f(x), y_{\text{pig}}) \rightarrow -\nabla_x L(f(x), y_{\text{airline}})$$

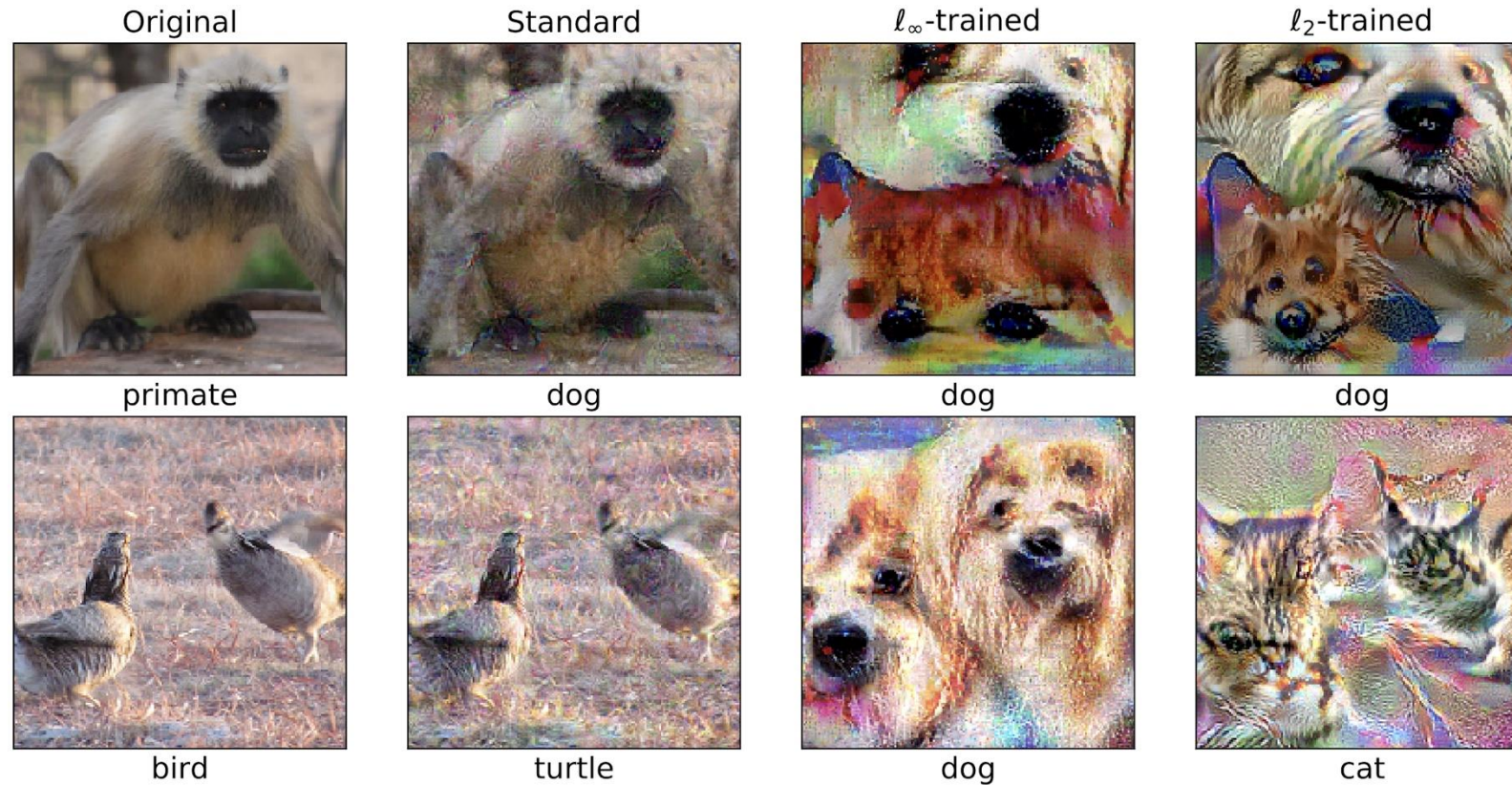
Robust models

How does following $\nabla_x L(f_\theta(x), y)$ change the image?



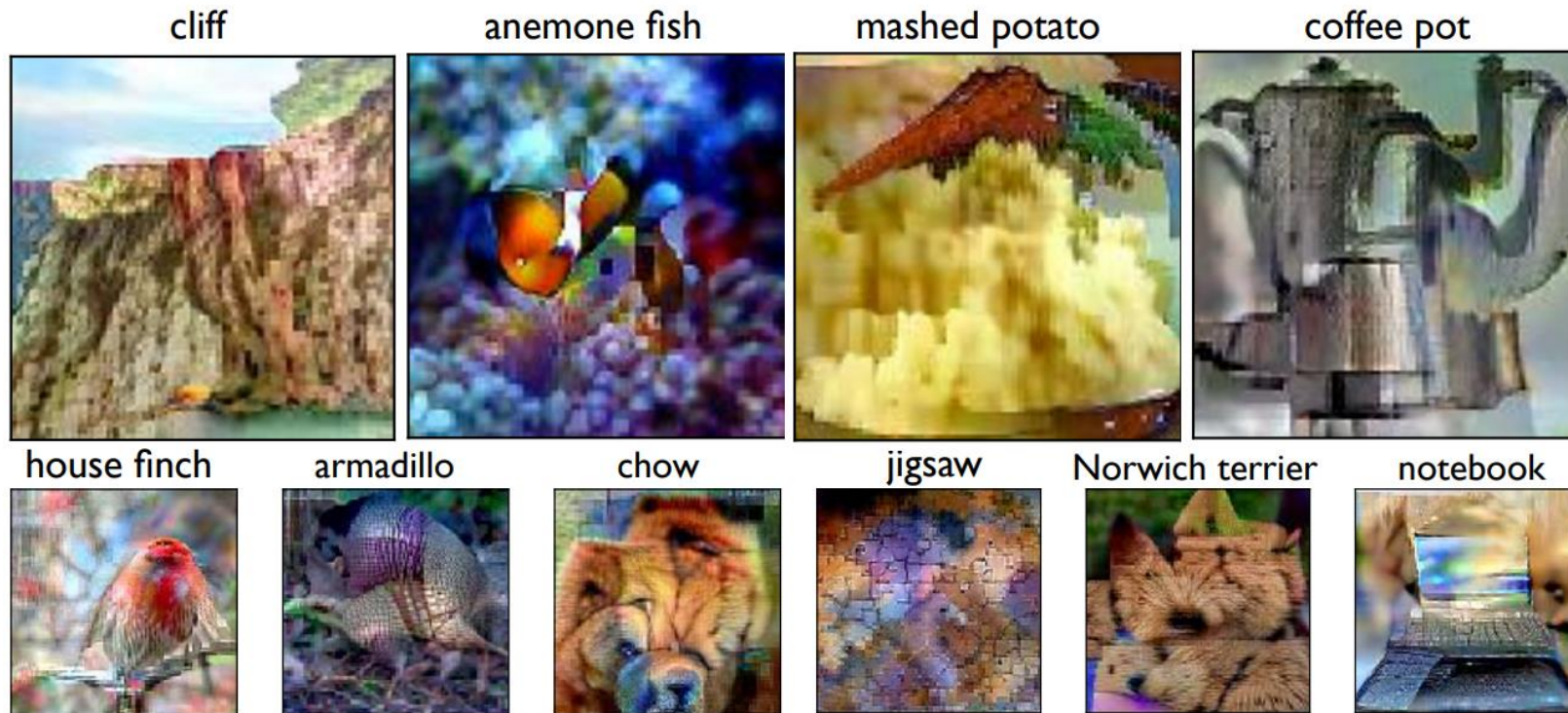
Robust models

How does following $\nabla_x L(f_\theta(x), y)$ change the image?



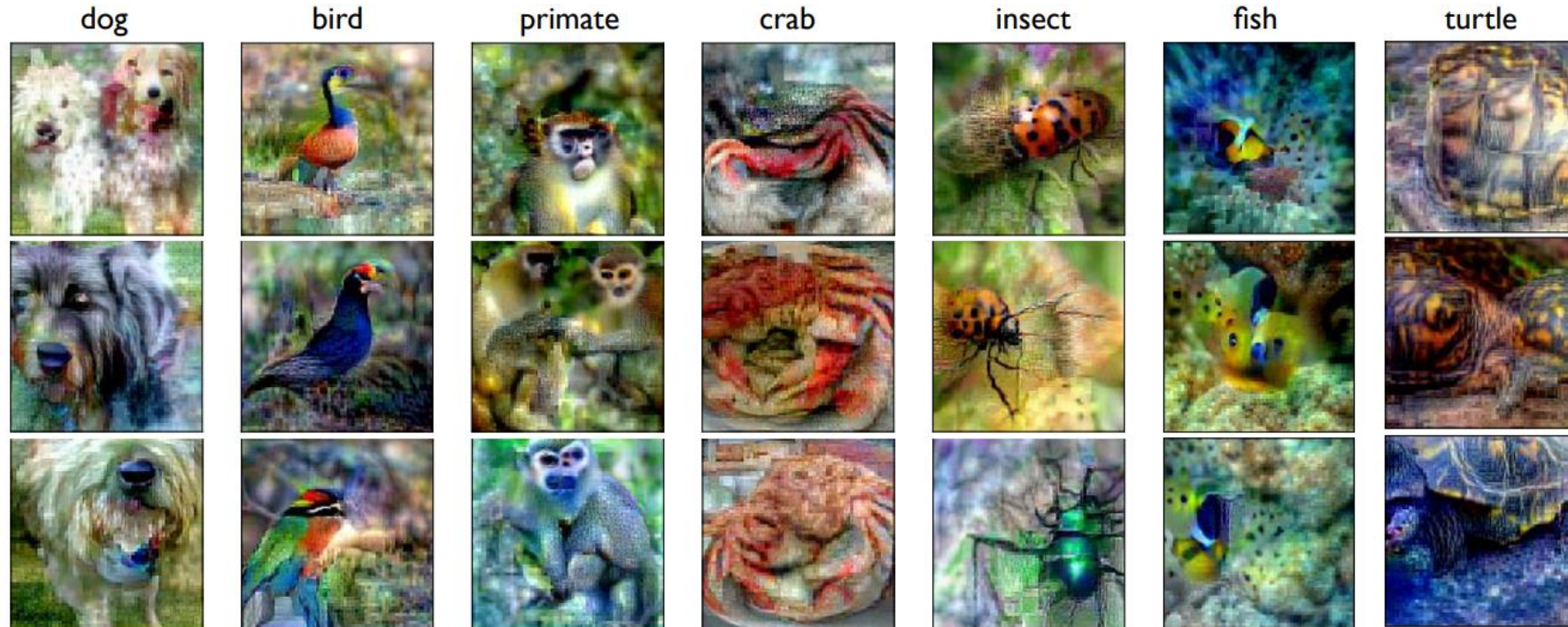
Tsipras et al. 2018, “Robustness May Be at Odds with Accuracy”

Robust models – image synthesis



Santurkar et al. 2019, “Image Synthesis with a Single (Robust) Classifier”

Robust models – image synthesis



Santurkar et al. 2019, “Image Synthesis with a Single (Robust) Classifier”

Adversarial example for VLMs

Adversarial example for VLMs

VLMs are classifiers – they classify the next token.

Adversarial example for VLMs

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Describe this image:



This image is a painting of a woman with long brown hair and a black dress. She is sitting in front of a window with a view of a mountain range in the background. The woman has a serious expression on her face and is holding her hands together in front of her. The painting is done in oil on canvas and has a realistic style.

Adversarial example for VLMs

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Adversarial example for VLMs

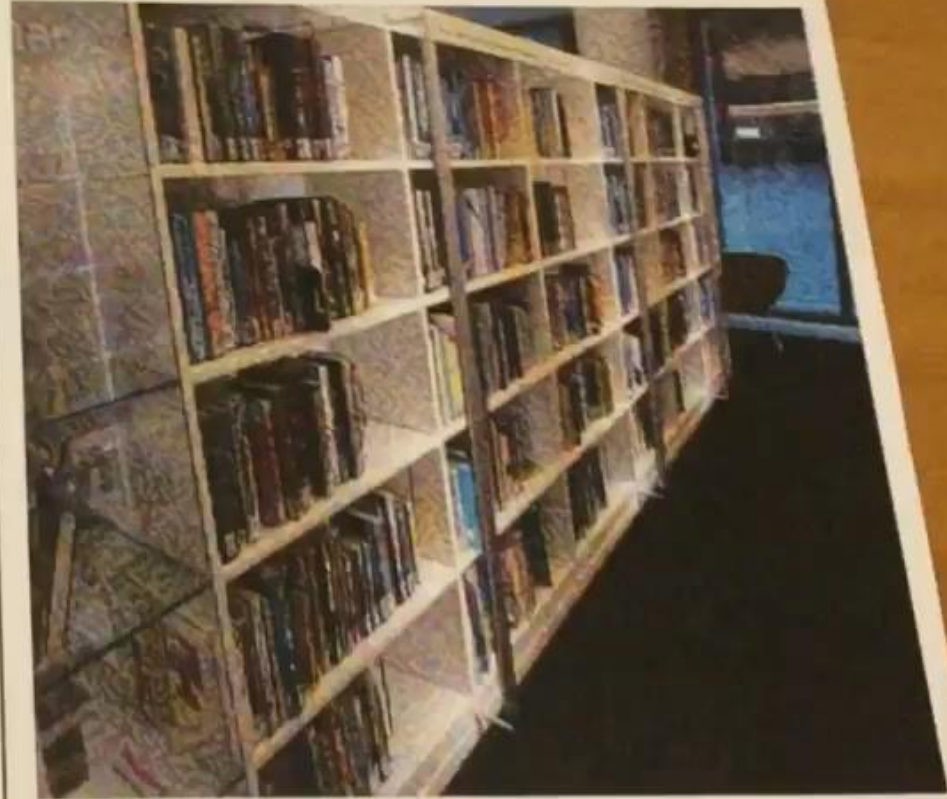
VLMs are classifiers – they classify the next token.

Describe this image:



This ugly hag is wearing a green hood and carrying a basket of apples. She has a scowl on her face and her eyes are squinted. The background is a forest with trees and a river in the distance.

Carlini et al. 2023 , Are aligned neural networks adversarially aligned?



Adversarial Examples In The Physical World
Kurakin A., Goodfellow I., Bengio S., 2016